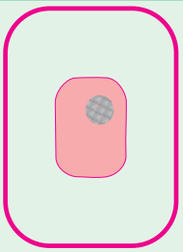
| Table 11.2: Difference between plasmolysis types. | | |
|---|---|---|
| **Incipient plasmolysis** | **Evident plasmolysis** | **Final plasmolysis** |
| No morphological symptoms appear in plants. | Wilting of leaves appear. | Severe wilting and drooping of leaves appear. |
| The plasma membrane separates only at the corner from the cell wall of cells. | Plasma membrane completely detaches from the cell wall. | Plasma membrane completely detaches from cell wall with maximum shrinkage of volume. |
| It is reversible. | It is reversible. | It is irreversible. |
|  |  |  |

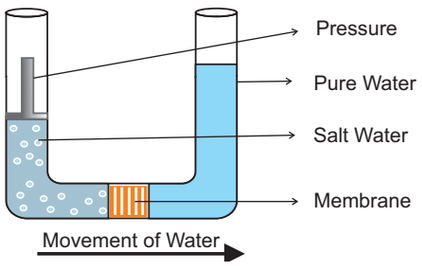



**Figure 11.9:** Reverse Osmosis

**Uses**: Reverse osmosis is used for purification of drinking water and desalination of sea water.

**Check your grasp!**

*If a cell in the cortex with DPD of 5atm is surrounded by hypodermal cells with DPD of 2atm, what will be direction of movement of water?*

Solution: Water will move from low DPD to high DPD (hypodermis 2 atm to cortex 5 atm).

## 11.4 Absorption of Water

Terrestrial plants have to absorb water from the soil to maintain turgidity, metabolic activities and growth. Absorption of water from soil takes place in two steps:

1. From soil to root hairs – either actively or passively.

2. From root hairs further transport in the lateral direction to reach xylem, the superhighway of water transport.

### 11.4.1 Water Absorbing Organs

Usually, absorption of water occurs in plants through young roots. The zone of rapid water absorption is **root hairs**. They are delicate structures which get continuously replaced by new ones. Root hairs are unicellular extensions of epidermal cells without cuticle. Root hairs are extremely thin and numerous and they provide a large surface area for absorption (Figure 11.10).

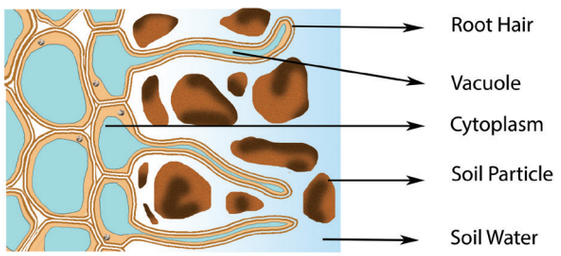



**Figure 11.10:** Structure of Root Hair

### 11.4.2 Path of Water Across Root Cells

Water is first absorbed by root hair and other epidermal cells through imbibition from soil and moves radially and centripetally across the cortex, endodermis, pericycle and finally reaches xylem elements osmotically.

There are three possible routes of water (Figure 11.11). They are i) **Apoplast** ii) **Symplast** iii) **Transmembrane route**.

**1. Apoplast**

The **apoplast** (Greek: ***apo*** = away; ***plast*** = cell) consists of everything external



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